Functions

1. Molybdenum is required for the function of metallo-enzyme xanthine oxidase.
2. Presence of small amount of molybdenum helps in the utilization of copper.

Sources



Figure 9.21 Sources of Molybdenum

Molybdenum is available in normal diets. Liver and kidney are good animal sources. Whole grains and legumes are vegetable sources.

Absorption and excretion

About 50 - 70 % of the ingested Molybdenum is readily absorbed in the small intestine. About half of the absorbed molybdenum is excreted in urine.

Deficiency

Molybdenum deficiency is rare in human beings.

9.3.10. Selenium

Selenium is an essential trace element for all species including humans.

Functions

1. Selenium is essential for normal growth and fertility.
2. Selenium is a component of the enzyme which reduces oxidized glutathione.
3. It is involved in immune mechanism and synthesis of ATP.
4. Vitamin E and Selenium prevent peroxidative damage to cellular and sub-cellular organelles and chiefly the membrane.
5. Selenium may be a cancer protective agent.
6. Supplements of selenium probably protect against toxic effects of heavy metals like mercury and silver.

Sources

Selenium is largely present in different foods. Selenium is present in foods of plant origin grown in selenium rich soils. Any normal diet can meet the daily requirement of selenium.

Absorption and excretion

The principal dietary forms of selenium,selenocysteine and selenomethionine are absorbed from gastrointestinal tract. Selenium homeostasis is achieved by removal of excess selenium in urine.

Deficiency

Selenium deficiency is very rarely seen in human beings. Deficiency of selenium

Unit 9.indd 249 21-05-2018 15:12:17